Cross-Asset Macro Volatility

Harvesting the Volatility Risk Premium

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Section 1: Overview & Foundations

Strategy Performance vs. S&P 500 Benchmark

1.1 Strategy Overview

We implement a **cross-asset relative value volatility strategy** to harvest the spread between implied and realized volatility - **the Volatility Risk Premium (VRP)** - across a diversified macroeconomic universe (Equities, Gold, Oil, FX, and High-Yield Credit).

	VRP MACRO STRATEGY	S&P 500 (SPY)
ANALYSIS PERIOD	2015 – 2025	2015 – 2025
ANNUALIZED RETURN	**14.03%**	12.95%
ANNUALIZED VOL	**9.21%**	17.56%
SHARPE RATIO	**1.47**	0.78
MAX DRAWDOWN	**-11.79%**	-33.72%
CORRELATION TO S&P	**0.147**	1.000
BETA VS S&P	**0.077**	**1.000**

1.2 Investment Universe

We construct our asset universe using **highly liquid ETFS** and their corresponding "VIX-style" **implied volatility indices**, which output annualized 30-day implied volatility (%).

Spot Assets

- Equities: S&P 500 (SPY)
- Metals / Safe Haven: Gold (GLD)
- Energy: Crude Oil (USO)
- Currencies: Euro FX (FXE)
- Credit: High-Yield Corporate Bonds (HYG)

Volatility Indices

- VIX
- GVZ
- OVX
- EVZ
- VIXHY

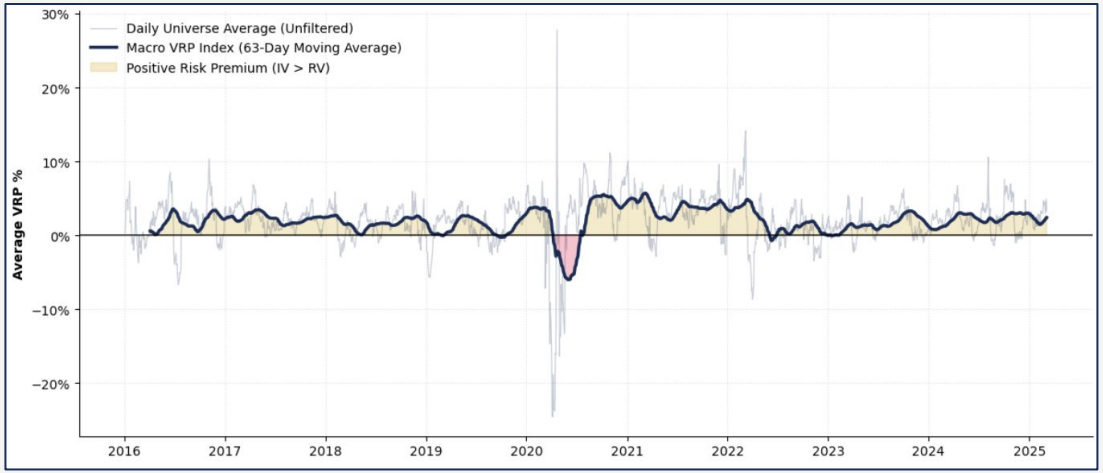
1.3 The Volatility Risk Premium

Volatility Risk Premium (VRP): The difference between forward-looking Implied Volatility and a 21-day rolling Realized Volatility window.

VRP = Implied Volatility - Realized Volatility

Volatility Indices (i.e., VIX)

21-Day Historical Lookback Calculated from Spot Assets



Left: *Equal-weighted average of VRP (IV - RV) across SPY, GLD, USO, FXE, and HYG.*

1.4 Theoretical Foundations

Delta-Hedged Gains and the Negative Market Volatility Risk Premium

Abstract

We investigate whether the volatility risk premium is negative by examining the statistical properties of delta-hedged option portfolios (buy the option and hedge with stock). Within a stochastic volatility framework, we demonstrate a correspondence between the sign and magnitude of the volatility risk premium and the mean delta-hedged portfolio returns. Using a sample of S&P 500 index options, we provide empirical tests that have the following general results. First, the delta-hedged strategy underperforms zero. Second, the documented underperformance is less for options away from the money. Third, the underperformance is greater at times of higher volatility. Fourth, the volatility risk premium significantly affects delta-hedged gains even after accounting for jump-fears. Our evidence is supportive of a negative market volatility risk premium.

Variance Risk Premia

ABSTRACT

We propose a direct and robust method for quantifying the variance risk premium on financial assets. We theoretically and numerically show that the risk-neutral expected value of the return variance, also known as the variance swap rate, is well approximated by the value of a particular portfolio of options. Ignoring the small approximation error, the difference between the realized variance and this synthetic variance swap rate quantifies the variance risk premium. Using a large options data set, we synthesize variance swap rates and investigate the historical behavior of variance risk premia on five stock indexes and 35 individual stocks.

Delta-Hedged Gains (Bakshi & Kapadia, 2003)

- Demonstrated that **implied volatility structurally overstates realized volatility** by proving that delta-hedged option buyers pay a persistent "insurance premium."
- Our strategy framework is designed to capture this structural overpricing.

Variance Risk Premia (Carr & Wu, 2009)

- Identifies the **VRP is a pervasive risk premium across all liquid asset classes** (Equities, Metals, Energy, and FX).
- Provides the mandate for our Diversified Macro Universe.

Section 2: Strategy Mechanics

2.1 Signal Generation & Trade Selection

To identify the daily relative value opportunity, we filter our macro universe through a three-stage process:

1. **Normalization:** The raw VRP for each asset is converted into a 252-day Rolling Z-Score.

2. **Ranking:** Assets are ranked from Highest Z-Score (Expensive) to Lowest Z-Score (Cheap).

3. **Selection:** The strategy isolates the two "extreme" tails of the distribution to construct the daily pair.

March 3, 2025

- **Oil (USO): +1.57**
- S&P 500 (SPY): +1.35
- High Yield (HYG): +0.71
- Gold (GLD): +0.24
- **Euro (FXE): -1.32**

Above: *Illustrative Z-Score Ranking*

2.2 Trade Execution

To capture the Volatility Risk Premium without taking directional market risk, we execute **At-The-Money (ATM) Straddles** on our selected tail assets.

Short Straddle:

- **Signal:** Highest Z-Score ("Expensive" Volatility).
- Sell 1 ATM Call + Sell 1 ATM Put
- Our primary VRP harvest - collects the structural premium paid by market hedgers.

Long Straddle:

- **Signal:** Lowest Z-Score ("Cheap" Volatility).
- Buy 1 ATM Call + Buy 1 ATM Put
- Our tail-risk insurance.

2.3 Daily P&L Attribution

For each asset in our relative value pair (Short and Long), the daily P&L of the straddle position is estimated as:

$$\text{Daily PnL} \approx \underbrace{\left(\tfrac{1}{2}\Gamma \cdot (\Delta S)^2\right)}_{\text{Gamma PnL}} + \underbrace{(\mathcal{V} \cdot \Delta\sigma)}_{\text{Vega PnL}} + \underbrace{\Theta}_{\text{Theta PnL}}$$

We approximate **Daily Option P&L** through a **second-order Taylor Series Expansion**

Framework adopted from Hull, J.C., "Options, Futures, and Other Derivatives"

Gamma: Realized Component

- Convex exposure to realized price shocks; captures Realized Volatility (RV).
- Spot price makes a violent move – **option prices** ⬆️ – **Benefits Buyer** (Long)

Vega: Implied Component

- Mark-to-market sensitivity to a 1% change in forward-looking Implied Volatility (IV).
- Implied Volatility rises – **option prices** ⬆️ – **Benefits Buyer** (Long)

Theta:

- Structural cost of holding options, daily time decay.
- Time passes (1 day goes by) – **option prices** ⬇️ – **Benefits Seller** (Short)

2.4 Black-Scholes: Application & Constraints

The Pricing of Options and Corporate Liabilities

Fischer Black
University of Chicago
Myron Scholes
Massachusetts Institute of Technology

If options are correctly priced in the market, it should not be possible to make sure profits by creating portfolios of long and short positions in options and their underlying stocks. Using this principle, a theoretical valuation formula for options is derived. Since almost all corporate liabilities can be viewed as combinations of options, the formula and the analysis that led to it are also applicable to corporate liabilities such as common stock, corporate bonds, and warrants. In particular, the formula can be used to derive the discount that should be applied to a corporate bond because of the possibility of default.

We extract portfolio partial derivatives (Greeks) via the **Black-Scholes-Merton (1973) framework.**

We do not use the model for absolute option pricing, as its core assumptions (**constant volatility** and continuous, **frictionless hedging**) contradict our structural thesis.

Our edge relies on monetizing stochastic volatility under discrete daily hedging constraints.

Section 3: Baseline Performance

3.1 Initial Backtest | Equal-Weight

We establish a **baseline performance** using a "naive" **equal-weight allocatio**n across all available trading days

Margin Mechanics:

- Trading ATM Straddles does not require the full notional value of the underlying asset.
- We post a fraction of the value as Initial Margin, leaving the remaining cash unencumbered to accrue the risk-free rate.

Prime Brokerage Framework:

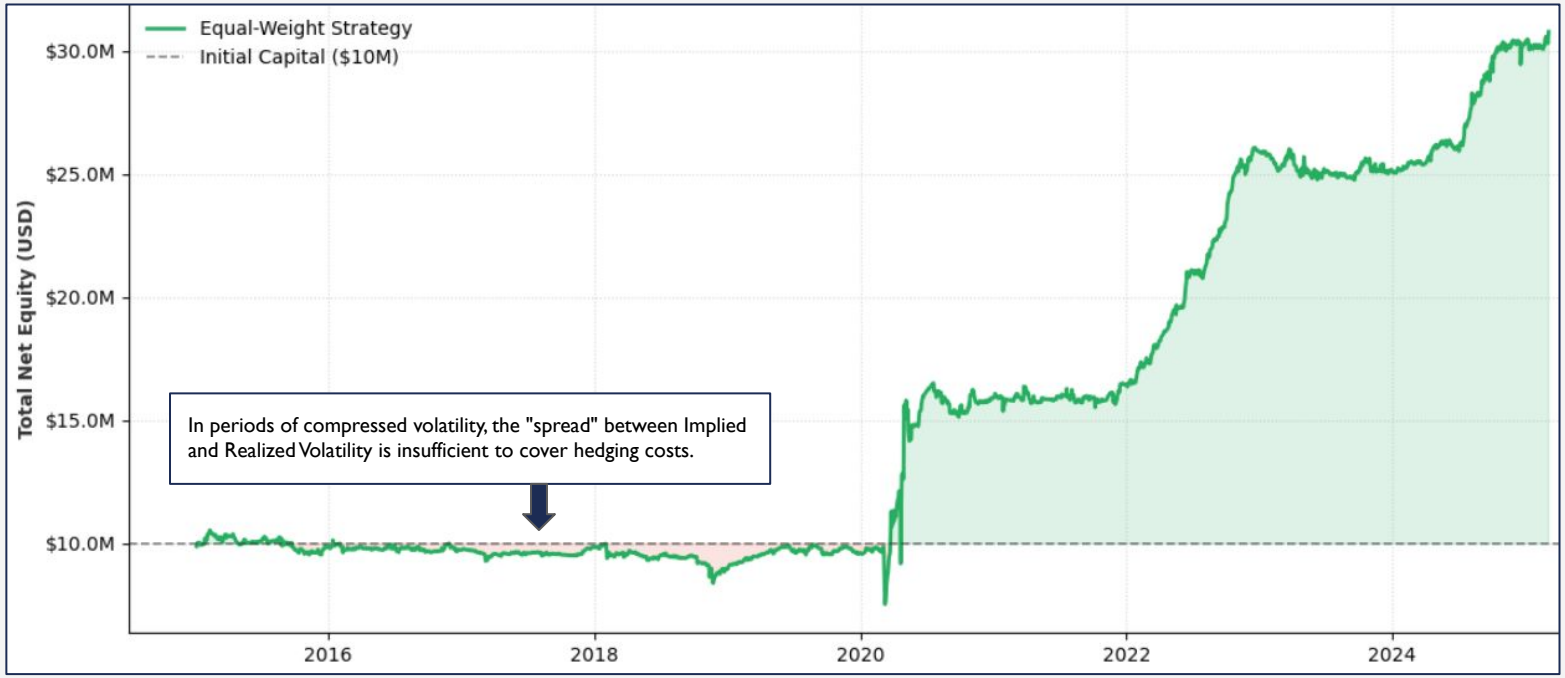
- We assume a **$10M** cash-collateralized account.
- A 2.0x leverage factor permits us to "write" (sell) and "buy" straddles representing **$20M** in underlying ETF notional value.

Friction & Funding:

- A 0.5 basis point (bps) penalty is applied to the estimated notional value of all shares traded. This simulates the slippage and market impact incurred during **daily delta-hedging**.
- A 50 basis point (bps) annualized rate is charged daily on the notional value of the short leg to simulate institutional **borrow costs** and **margin maintenance.**

3.2 Performance Metrics | Equal-Weight

	ANN RETURN	ANN VOL	SHARPE	MAX DD	ACTIVE DAYS %	TRADING DAYS	GROSS OPT PNL	CASH YIELD	HEDGE COSTS	BORROW COSTS	NET PNL	FINAL EQUITY
VRP Equal-Weight	11.72%	16.94%	0.75	-28.56%	100.0%	2,557	$23,187,726	$4,004,892	-$5,621,195	-$782,434	**$20,788,989**	**$30,788,989**



3.3 Historical Drawdowns | Equal-Weight

Drawdown profile measures the percentage (%) decline from the portfolio's previous high-water mark.

The sharp drawdown in early 2020 highlights the equal-weight strategy's **vulnerability to correlated volatility spikes** across asset classes.

Section 4: Risk-Parity Weighting

4.1 Inverse-Volatility Weighting | Risk-Parity

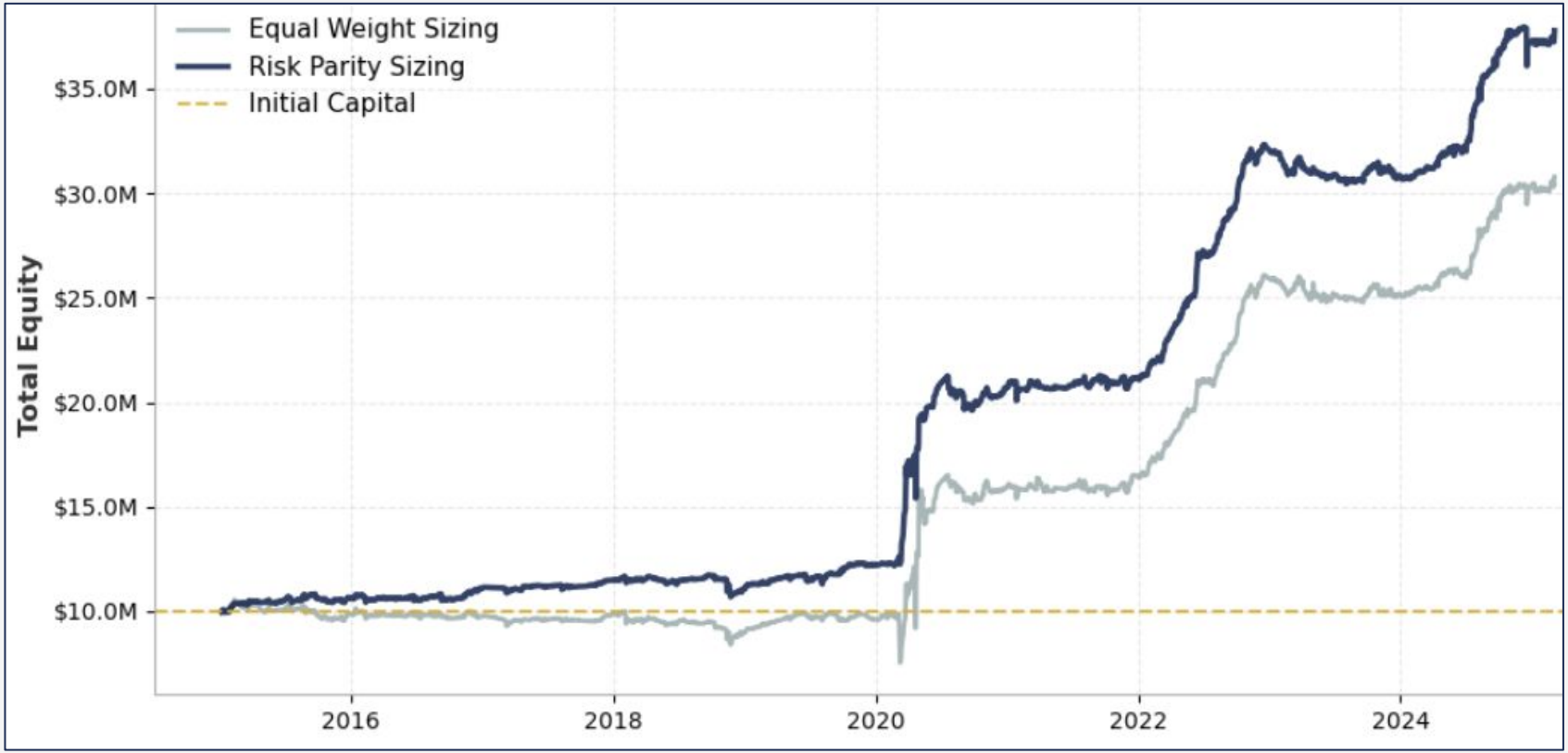
Our initial backtest allocated capital equally across the two selected assets each day, ignoring their risk profiles.

The **Inverse-Volatility (Risk Parity)** weighting scheme equalizes the risk contribution across the portfolio:

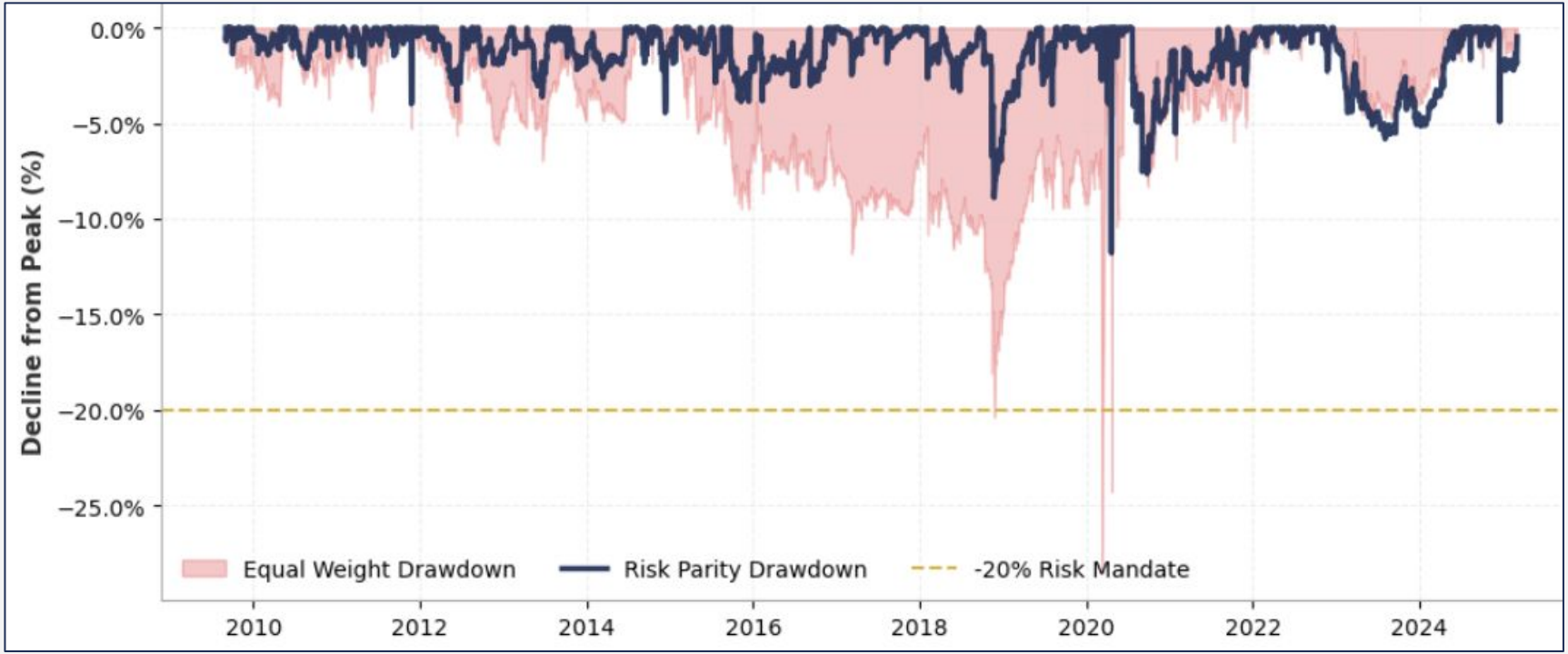
- We **scale the notional exposure** of each leg **based on its Realized Volatility (RV).**
- Low-volatility assets receive higher notional weight.
- High-volatility assets receive lower weight.

	ANN RETURN	ANN VOL	SHARPE	MAX DD	ACTIVE DAYS %	TRADING DAYS	GROSS OPT PNL	CASH YIELD	HEDGE COSTS	BORROW COSTS	NET PNL	FINAL EQUITY
EQUAL WEIGHT	11.72%	16.94%	0.75	-28.56%	100.0%	2,557	$23,187,726	$4,004,892	-$5,621,195	-$782,434	**$20,788,989**	**$30,788,989**
INVERSE VOL.	14.00%	9.21%	1.47	-11.79%	100.0%	2,557	$31,222,145	$4,925,033	-$7,219,997	-$1,144,494	**$27,782,688**	**$37,782,688**

4.2 Growth Comparison | Risk-Parity vs. Equal-Weight



4.3 Drawdown Analysis | Risk-Parity vs. Equal-Weight



Section 5: Benchmark Comparisons

5.1 Volatility Benchmarks | Risk-Adjusted Returns

To justify its role in an institutional portfolio, our strategy must outperform other passive volatility instruments.

	STRATEGY	EQUITY (SPY)	SHORT VOL (SVXY)	LONG VOL (VXX)	TREND (DBMF)
ANALYSIS PERIOD	2020 – 2025	2020 – 2025	2020 – 2025	2020 – 2025	2020 – 2025
ANNUALIZED RETURN	**25.00%**	16.14%	16.17%	-26.13%	6.31%
ANNUALIZED VOL	**11.35%**	20.81%	39.05%	75.31%	12.85%
SHARPE RATIO	**1.97**	0.72	0.38	-0.40	0.48
MAX DRAWDOWN	**-11.79%**	**-33.72%**	**-62.19%**	**-99.08%**	**-20.39%**

Analysis period standardized to 2020-2025 to account for the May 8, 2019 launch of DBMF (Managed Futures ETF).

5.2 Volatility Benchmarks | Correlation

	Strategy	S&P 500	Short Vol (SVXY)	Long Vol (VXX)	Trend (DBMF)
Strategy	1.000	0.113	0.113	-0.125	0.054
S&P 500	0.113	1.000	0.735	-0.713	0.157
Short Vol (SVXY)	0.113	0.735	1.000	-0.980	0.254
Long Vol (VXX)	-0.125	-0.713	-0.980	1.000	-0.265
Trend (DBMF)	0.054	0.157	0.254	-0.265	1.000

The Risk-Parity approach maintains **minimal correlation** to both broad equity markets (S&P 500) and passive volatility/trend-following benchmarks.

Section 6: Conclusion

6.1 The Case for Cross-Asset Volatility Carry

Our analysis confirms that the Volatility Risk Premium (VRP) remains a robust and harvestable structural risk premium across diversified asset classes.

- **Risk-Adjusted Returns:** The strategy delivered a 1.47 Sharpe Ratio (2015–2025), significantly outperforming the S&P 500 (0.78) and passive volatility instruments.
- **Low Correlation:** With a Beta of 0.077 and a correlation of 0.113 to the S&P 500, the strategy acts as a powerful diversifier.
- **Tail Resilience:** The strategy's Long position mitigated drawdown to just -11.79% during major market shocks.